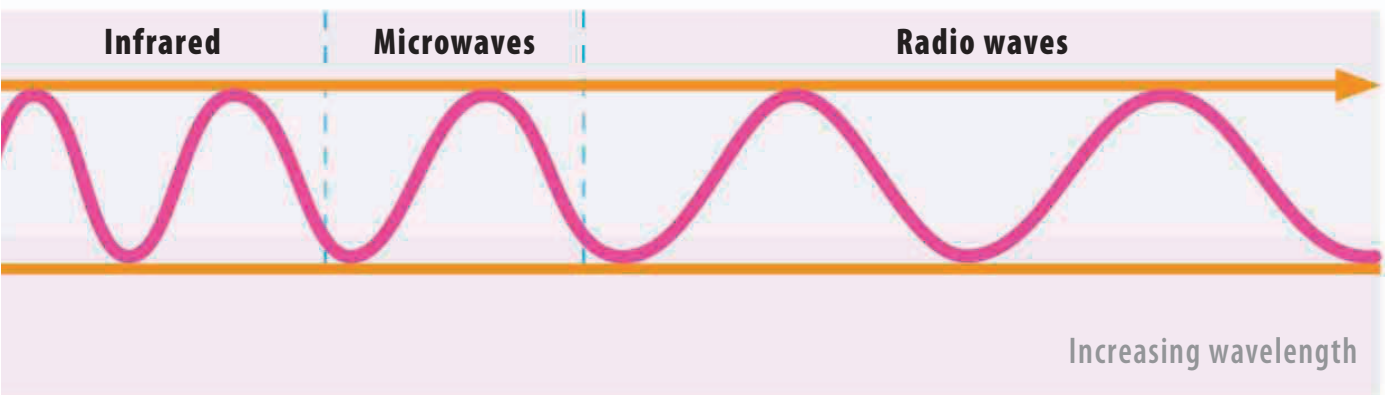
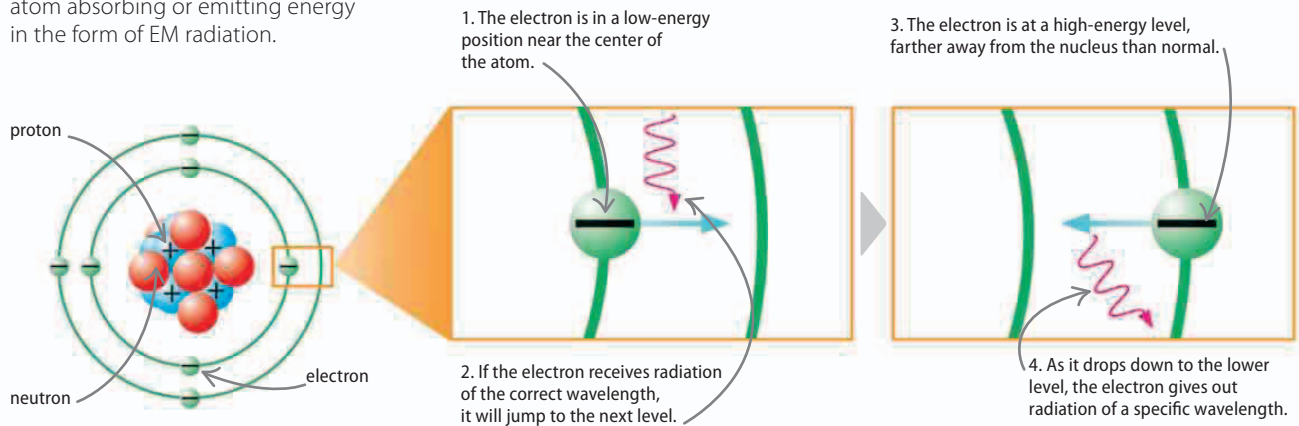


The source of EM radiation

EM radiation is associated with the force that holds electrons in place around atoms (see pages 168–169). However, the electrons can move around, jumping between higher and lower energy levels, or shells. These changes result in the atom absorbing or emitting energy in the form of EM radiation.



Infrared

Infrared means "below red," and has a lower frequency and longer wavelength than visible red light. We experience infrared waves as heat, and can see it at work in heaters, grills, and toasters. It is also used in television remote controls and in fiber optics.

Microwaves

This band of wavelengths is used in many types of personal communications, including mobile phones, wi-fi, and Bluetooth, as well as in microwave ovens. It is also used in radar technology, as a way of locating airplanes and ships.

Radio waves

Radio waves are the longest in the spectrum. They are used to transmit radio and TV signals around Earth. Television uses higher frequencies than radio. Radio waves from space can be picked up using radio telescopes and used to study the Universe.